# Yulong Liu

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## Research Interests

**AI methods & discovery:** Differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, and AI for Science and scientific discovery.

**Engineering:** Homogenization and multiscale modeling, coupled multiphysics simulation, and subsurface energy systems, including geothermal reservoirs and hydraulic fracturing.

## Education

**Ph.D.** in Earth Science – Cornell University 2024 – Present
*Minors: Computer Science and Scientific Computing*
**B.S.** in Mining Engineering – Northeastern University (CN) 2024

## Research Experience

**Ph.D. Candidate** – Dept. of Earth and Atmospheric Sciences, Cornell University Aug. 2024 – Present
*Advisor: Chloé Arson* *Ithaca, NY*

- Develop coupled thermal, hydraulic and mechanical models in MOOSE for geothermal reservoirs.
- Build joint temperature–pressure neural operators for geothermal forecasting, uncertainty propagation, Bayesian reservoir inference and monitoring design.
- Study differentiable programming and PDE neural solvers, integrating physics-informed networks, implicit neural representations and neural operators with numerical simulation.
- Investigate parameterization of theoretical homogenization and multiscale transport in porous media, using scientific machine learning to discover interpretable effective permeability laws.
- Advance agentic LLMs for science; lead a cross-institution project on physics-grounded verification that generated multiphysics simulation code represents the intended physics.

**Undergraduate Researcher** – Northeastern University (CN) Sep. 2020 – Jun. 2024
*B.S. in Mining Engineering* *Shenyang, China*
**Thesis:** A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning

- Developed a joint deep-learning model for microseismic signal classification and arrival-time picking, integrating tasks typically handled in separate workflows.
- Conducted experimental and computational research on hydraulic fracturing, triaxial rock testing, and tunnel boring machines; received the Outstanding Thesis Award and National Scholarship.

## Research Projects and Funding

**Joint Operator Learning and Digital Twins for Enhanced Geothermal Systems** 2025 – Present
*Coupled reservoir simulations, joint temperature–pressure forecasting and observation-based permeability inference for the Cornell geothermal case study.*

- **Cornell Atkinson Academic Venture Fund:** *AI-aided Drilling for Deep Geothermal Development: Cornell Site Case Study*. PI: C. Arson; Co-PI: Y. Yang. 2025–2027.
- **NSF R2I2, Award #2517787:** *Enhancing Northeast Resilience through Innovations for Deep Geothermal Heat Direct Use*. PI: C. Arson; Co-PIs: S. Olmstead and J. Tester. 2025–2027.
- **Cornell Engineering SPROUT-energy:** *Mathematical Modeling Framework for Enhanced Geothermal Systems Digital Twins*. 2026.
- **NSF ACCESS project (EES260150):** *Learned Uncertainty-Propagation and World Models for Enhanced Geothermal Reservoirs*. PIs: **Y. Liu** and C. Arson. 1.5M ACCESS Credits (equivalent to $37.5K). 2026–2027.
- **Empire AI Alpha+Beta:** *AI-aided Computational Geomechanics*. PIs: Y. C. Han, **Y. Liu**, A. Tristani, and C. Arson. 5.5K GPU hours (equivalent to $2.75K). 2026.

- **Simons Foundation, Utilizing the Full Power of Empire AI** (proposal submitted, 2027). PI: C. Arson. **Role:** Proposal Lead and Scientific Co-Lead.
  - Invited by the faculty PI to serve as Co-PI; formal Co-PI designation was precluded by institutional and sponsor eligibility requirements for graduate students.
  - Led the scientific conceptualization, proposal development, and research design.
  - Responsible for leading the proposed research program on physics-informed foundation world models and computational multiphysics.

**Theoretical Homogenization of Multiscale Porous Media Flow by Machine Learning and Foundation Models** 2025 – Present

*Homogenization theory and scientific machine learning for interpretable permeability laws.*

- **Cornell Engineering SPROUT:** *Optimizing Metrics for Transport Homogenization: Subsurface Energy Systems Predictive Models.* Aug. 2025 – Jul. 2026.
- **NSF ACCESS project (EES260208):** *Homogenization and Machine Learning for Multiscale Porous Media Flow.* PIs: **Y. Liu** and C. Arson. 1.5M ACCESS Credits (equivalent to $37.5K). 2026–2027.
- **Simons Foundation, Utilizing the Full Power of Empire AI** (proposal in preparation). PI: Y. Yang; Co-PI: C. Arson. **Role:** Contributing Author. Topic: scientific foundation models (full title withheld).

**Groundwater Digital Twin** Proposed

- **NASA ROSES-25 F.5 FINESST** (under review). Future Investigator: **Y. Liu**; PI: C. Arson. Full title withheld.

## Publications

### Selected Papers

- **Y. Liu**, J. Botvinick-Greenhouse, Y. Yang, C. Arson. *Operator Learning Surrogate Modeling of Hydraulically Fractured Geothermal Injection–Production Systems: A Cornell Case Study.* ARMA US Rock Mechanics/Geomechanics Symposium, Tucson, AZ, 2026.
- **Y. Liu**, C. Arson. *A Physics-Informed Neural Network for Pressurized Cavities of Arbitrary Shape in Heterogeneous Rock. Rock Mechanics and Rock Engineering* (2026). doi:10.1007/s00603-026-05882-5.
- M. Belachew, **Y. Liu**, J. D. Frost, C. Arson. *Numerical Assessment of Plasticity Development and Energy Expenditure of Ant-Like Microtunnelling. Tunnelling and Underground Space Technology*, 172, 107501, 2026.
- **Y. Liu**†, C. Arson. *Physics-Informed Neural Network Surrogate Modeling of Pressurized Cavity in Homogeneous and Bilayered Media.* ARMA US Rock Mechanics/Geomechanics Symposium, D022S018R006, 2025.
- **Y. Liu***,†, Z. Song*, et al. *Your Simulation Runs but Solves the Wrong Physics: PDE-Grounded Intent Verification for LLM-Generated Multiphysics Simulation Code.* arXiv:2605.09360. Under review at NeurIPS 2026.
- Z. Song, Y. Li, **Y. Liu**†. *Prefix-Safe Bayesian Belief Tracking for LLM Reasoning Reliability: Separating Calibration from Ranking.* arXiv:2605.27712. Under review at ACL ARR 2026.

### Thesis

- **Y. Liu**. *A Joint Model for Microseismic Signal Classification and Arrival-Time Picking Based on Deep Learning.* Undergraduate thesis, Northeastern University, 2024.

* Co-first author † Corresponding author

## Conferences and Presentations

**Invited participant, CHiMaD Phase Field Methods XXI** Upcoming
*Center for Hierarchical Materials Design (CHiMaD)*

**Poster presentation, 60th US Rock Mechanics/Geomechanics Symposium (ARMA)** Jun. 2026
*American Rock Mechanics Association, Tucson, AZ, USA*

**Panelist, CHiMaD Phase Field Methods XX** May 2026
*Center for Hierarchical Materials Design, Northwestern University, Evanston, IL, USA*

**Poster presentation, Cornell CEE Graduate Research Symposium** Apr. 2026
*Cornell University, Ithaca, NY, USA*

**Invited seminar: "Physics-Informed AI Optimization of Heterogeneous Rock Systems"** Dec. 2025
*Southwest Petroleum University seminar series, Chengdu, China*

**Poster presentation, 59th US Rock Mechanics/Geomechanics Symposium (ARMA)** Jun. 2025
*American Rock Mechanics Association, Santa Fe, NM, USA*

## Fellowships and Awards

**Estwing Hammer Award** – Most Outstanding EAS Graduate Student of 2024, Cornell University 2025
**Cornell University Travel Grant**, Cornell University 2025, 2026
**Outstanding Thesis**, Northeastern University 2024
**National Scholarship**, Ministry of Education, China 2024
**First Class Scholarship**, Northeastern University 2023
**Autumn Scholarship**, Northeastern University 2022
**Golden Seed Scholarship**, Northeastern University 2022

## Academic Service

- Session Chair, GS03 Computational Geomechanics, Coupled Thermo-Hydro-Mechanical-Chemical (THMC) Processes, and Next-Generation Modeling, Geothermal Track, *61st U.S. Rock Mechanics/Geomechanics Symposium (ARMA 2027)*, Austin, TX, June 2027
- Reviewer, *Rock Mechanics and Rock Engineering*
- Reviewer, *Underground Space*
- Reviewer, *International Conference on Artificial Neural Networks*
- Reviewer, *Geothermal Rising Conference*

## Skills

**Programming & Scientific Computing:** Python, C++, R, MOOSE finite element framework

**Modeling & Simulation:** finite element modeling, coupled multiphysics simulation, poromechanics, computational geomechanics, CAD (e.g., AutoCAD, SolidWorks)

**Subsurface & Reservoir:** geothermal reservoirs, hydraulic fracturing, injection–production modeling, homogenization and multiscale transport in porous media

**Scientific Machine Learning:** differentiable programming, hybrid neural PDE solvers, uncertainty-aware scientific AI, AI for Science and scientific discovery; physics-informed networks, neural operators, and implicit neural representations

**LLMs for Simulation:** using large language models to accelerate simulation-software workflows and to verify the physical correctness of generated simulation code

**Experimental Methods:** rock mechanics testing (e.g., triaxial compression), laboratory instrumentation

**Currently Learning:** phase-field modeling (fracture/multiphysics), multiphase flow in porous media

**Scientific Writing:** LaTeX / Overleaf

## Professional Memberships

- Member, American Rock Mechanics Association (ARMA), 2025–Present
- Member, International Society for Rock Mechanics and Rock Engineering (ISRM), 2025–Present
- Member, Chinese Society for Rock Mechanics and Engineering (CSRME), 2024–Present
- Member, Geothermal Rising, 2025–Present